

“Linear Regression EDA”

Key Visualizations & Insights

1. Correlation Heatmap

Shows relationships between all encoded numerical variables.

Key Insights:

- **Smoking status (smoker_yes)** has the strongest positive correlation with charges.
→ Smokers pay significantly higher insurance costs.
 - **BMI** and **age** also show moderate positive correlation with charges.
→ Older individuals with higher BMI tend to have higher medical expenses.
 - Regions have near-zero correlation.
→ Location does NOT strongly affect insurance cost.
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2. Actual vs Predicted Plot

Shows how close model predictions are to actual insurance charges.

Insights:

- The model captures the **overall upward trend**, meaning predictions increase when actual values increase.
 - Some scatter remains → expected because charges also depend on non-linear patterns.
 - Good separation between smokers and non-smokers clusters was observed indirectly.
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3. Residual Plot

Residual = actual – predicted.

Insights:

- Residuals are spread around 0 but not perfectly centered.
 - Non-constant variance appears (wider spread for higher predictions).
→ Suggests heteroscedasticity (common in medical cost data).
 - A polynomial or regularized model may improve performance.
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● 4. Residual Distribution

Shows how errors are distributed.

Insights:

- Residuals are roughly normal but skewed to the right.
→ Model underestimates some high charges.
 - Long right tail shows **outliers** — mainly smokers with very high bills.
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Model Performance

Metric	Value
MAE	4181.19
MSE	33,596,915
RMSE	5796.28
R ² Score	0.78

Interpretation:

- The model explains **78% of variance**, which is strong for real-world medical cost data.
- On average, predictions are off by **\$4,181**, acceptable because actual charges range from **\$1,100 to \$63,700**.
- RMSE indicates some very large outliers (mostly smokers).